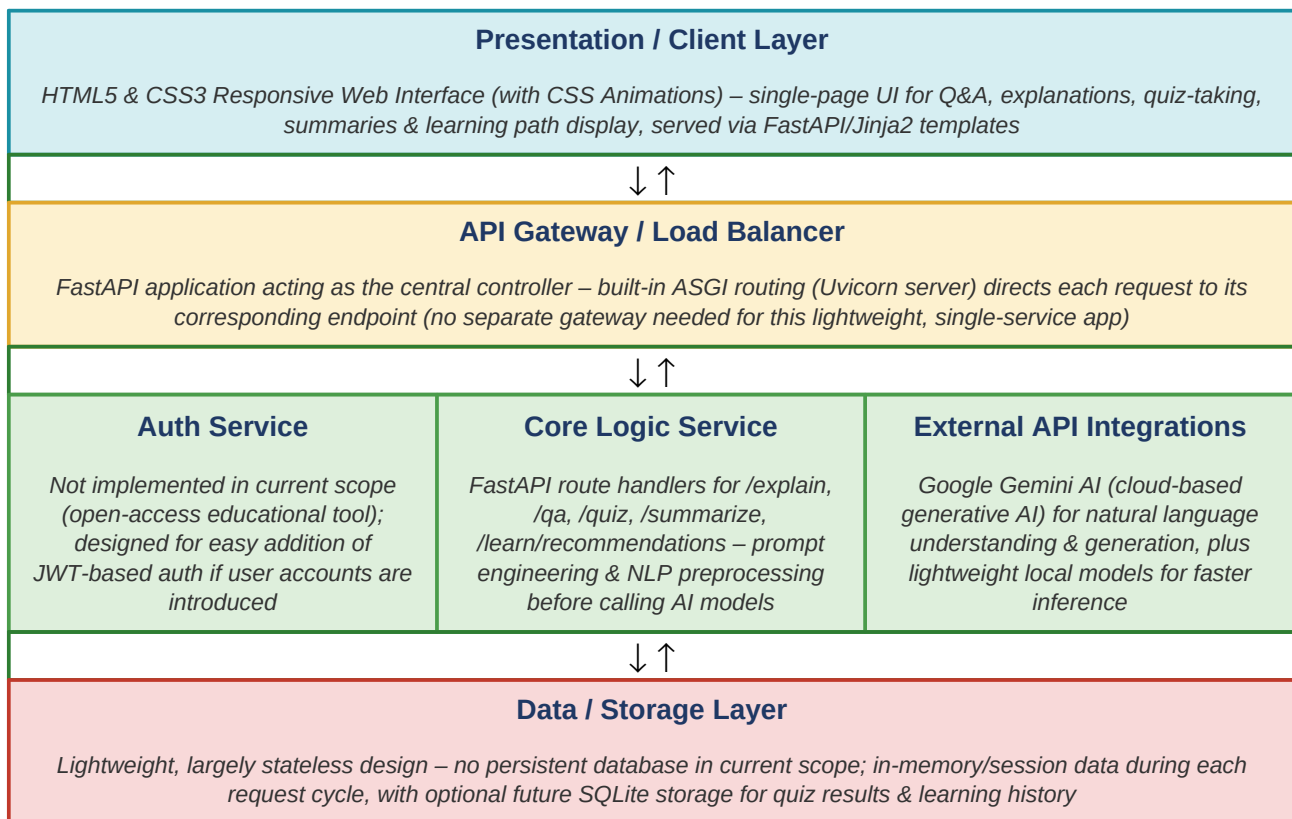


Solution Architecture

Date	June 26, 2026
Team ID	SWTID-2026-6095
Project Name	EduGenie
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	User-facing interface where students, self-learners, and educators enter questions, select a topic for explanation, generate a quiz, submit text for summarization, or request a learning path, and view the AI-generated results.	HTML5, CSS3 (animations), Jinja2 templates rendered via FastAPI
API Gateway	Routes each incoming user request to the appropriate backend endpoint based on the selected feature (Q&A, explanation, quiz, summary, or learning path); handled directly by FastAPI's built-in ASGI routing.	FastAPI, Uvicorn (ASGI server)
Microservices	Core logic that processes each request type using prompt engineering and NLP preprocessing, then calls the generative AI model to produce explanations, answers, quizzes, summaries, or personalized learning roadmaps.	Python, FastAPI routers, Prompt Engineering, NLP, Google Gemini AI, AI/ML Inference
Database	Lightweight, largely stateless data handling; no persistent database in the current scope, with results held only for the duration of a request/session.	In-memory Python session data (current scope); optional future SQLite for quiz results & learning history